Assignment 2 Report : RB Trees

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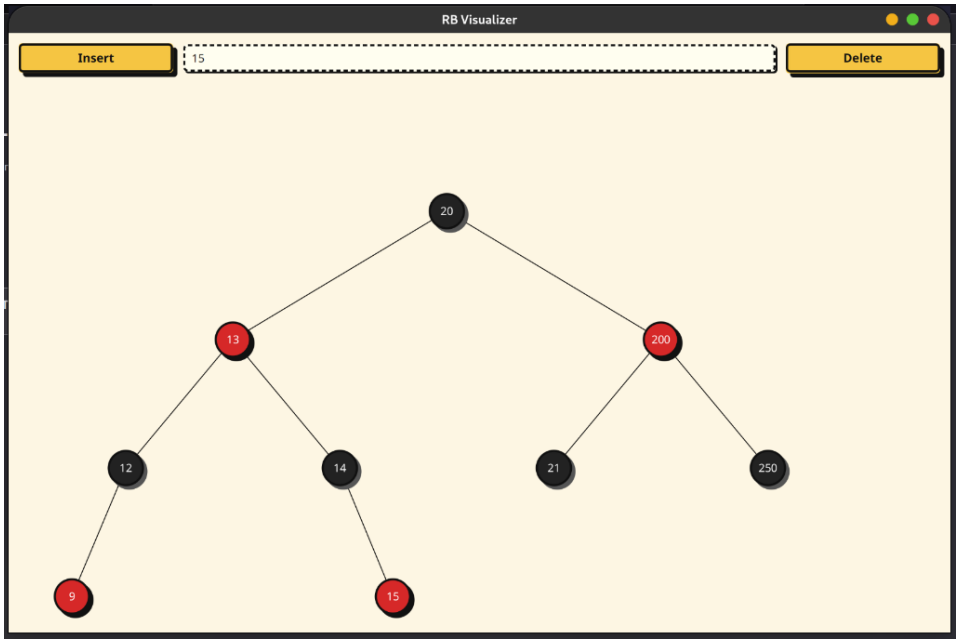
1 Visualization UI



2 CLI Performance Analysis

```
Enter the size of the array (Max is 1e7)
----------
1e6
Enter the array type (1) Sorted, (2) Reversed, (3) Random, (4) Partly Sorted
----------
4
Enter the Sorted percentage
----------
80
Enter the tree type (1) Simple Tree.BST, (2) Red-Black Tree.BST
----------
2

RB Tree Performance Analysis
-----------------------------
|Algorithm     |      Time(ms)|
|TreeSort      |         360ms|
|MergeSort     |         361ms|
```

3 Benchmark Data (run3.txt Summary)

3.1 10% Randomization

- **RBT Mean Time:** 26.49ms
- **Naive BST Mean Time:** 20775.69ms
- **Speed Up:** 784.20x

- **RBT Height:** 28 vs **Naive BST Height:** 56883

3.2 5% Randomization

- **RBT Mean Time:** 21.17ms
- **Naive BST Mean Time:** 20806.77ms
- **Speed Up:** 982.79x
- **RBT Height:** 28 vs **Naive BST Height:** 59982

3.3 1% Randomization

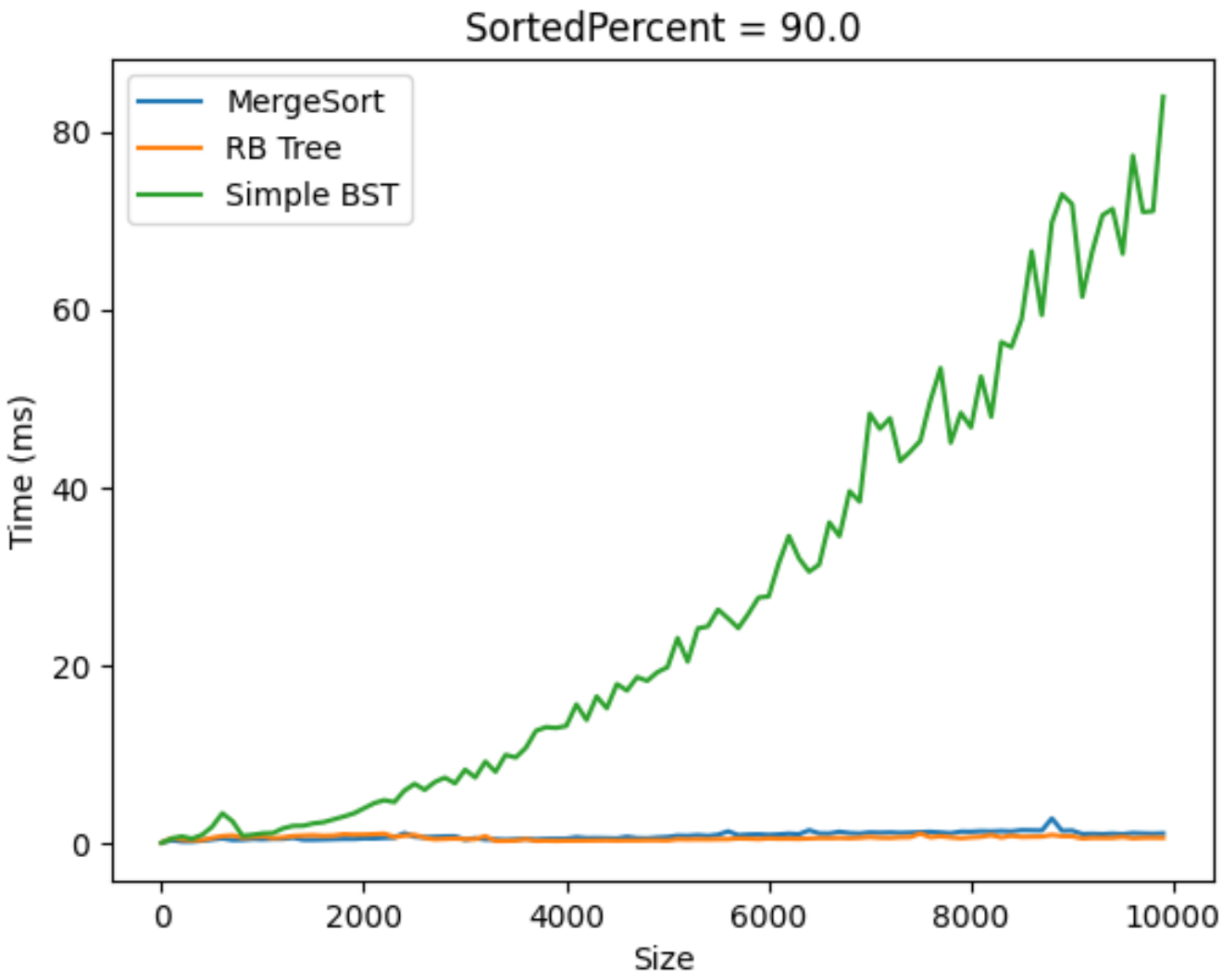
- **RBT Mean Time:** 21.18ms
- **Naive BST Mean Time:** 22116.23ms
- **Speed Up:** 1043.97x
- **RBT Height:** 28 vs **Naive BST Height:** 62566

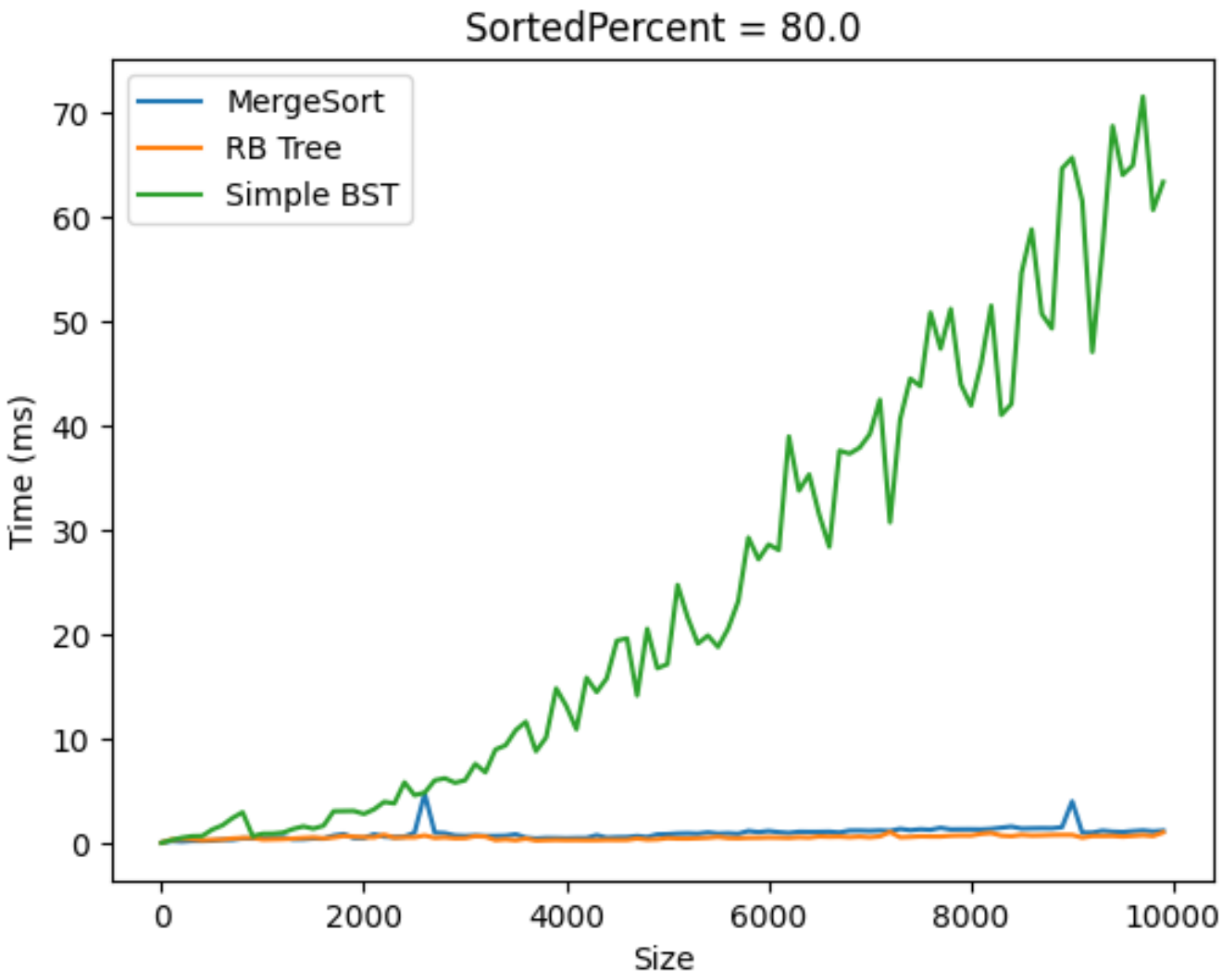
4 Sample Runtime CSV Data

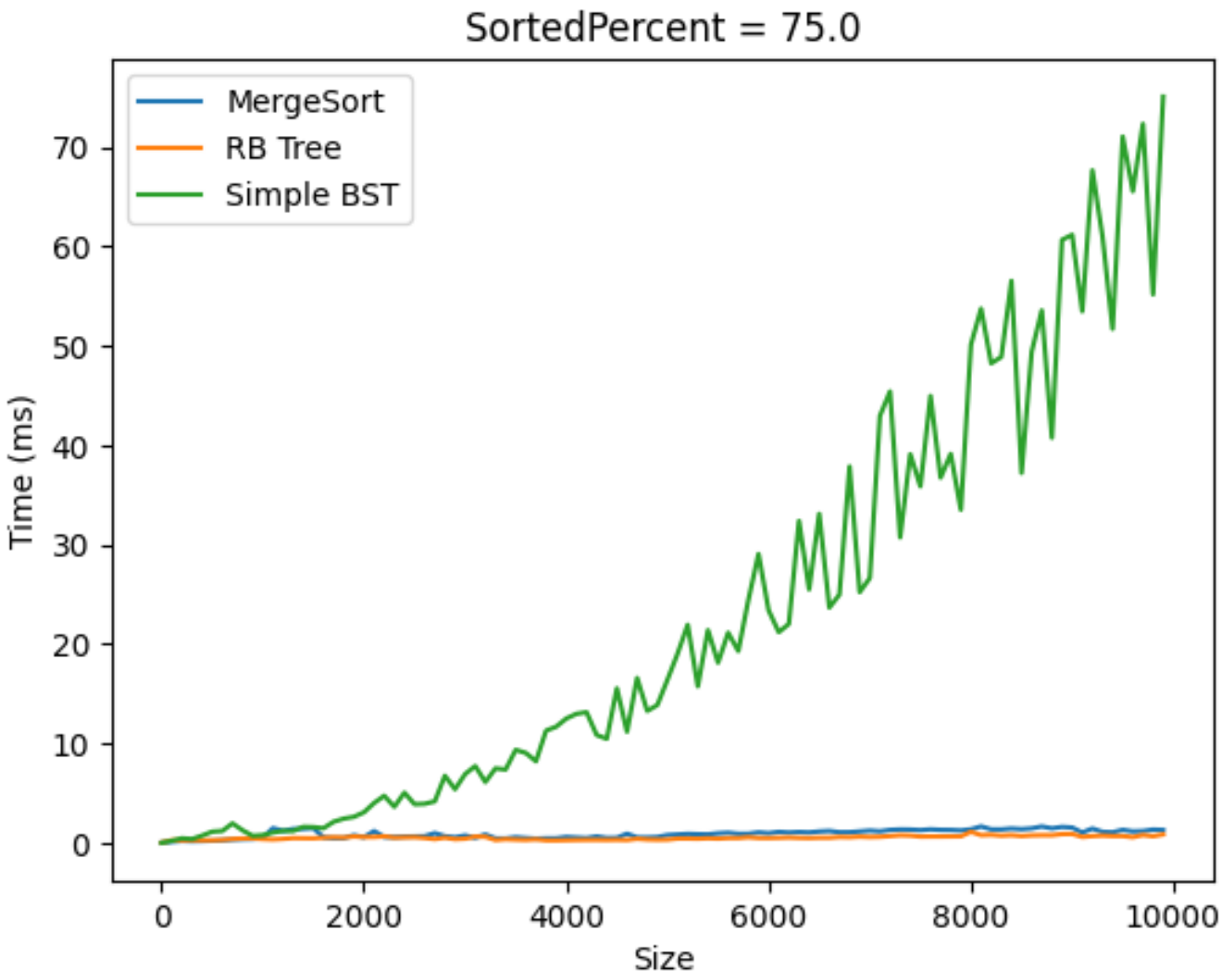
Algorithm	Size	Percent	Normal (ms)	RB (ms)	BST (ms)
MergeSort	1	80.0	0.018	0.029	0.006
MergeSort	101	80.0	0.181	0.310	0.283
MergeSort	201	80.0	0.130	0.366	0.498
MergeSort	301	80.0	0.195	0.358	0.642
MergeSort	401	80.0	0.190	0.264	0.662
MergeSort	501	80.0	0.213	0.298	1.280
MergeSort	601	80.0	0.300	0.326	1.682
MergeSort	701	80.0	0.304	0.428	2.382
MergeSort	801	80.0	0.439	0.480	2.937
MergeSort	901	80.0	0.419	0.493	0.650
MergeSort	1001	80.0	0.428	0.319	0.881
MergeSort	1101	80.0	0.514	0.345	0.889
MergeSort	1201	80.0	0.545	0.368	0.983
MergeSort	1301	80.0	0.347	0.421	1.337
MergeSort	1401	80.0	0.356	0.480	1.585
MergeSort	1501	80.0	0.429	0.519	1.369
MergeSort	1601	80.0	0.409	0.439	1.647
MergeSort	1701	80.0	0.705	0.523	3.034
MergeSort	1801	80.0	0.845	0.669	3.048
MergeSort	1901	80.0	0.473	0.594	3.063
MergeSort	2001	80.0	0.472	0.550	2.775
MergeSort	2101	80.0	0.829	0.535	3.219
MergeSort	2201	80.0	0.602	0.783	3.930
MergeSort	2301	80.0	0.557	0.497	3.778
MergeSort	2401	80.0	0.599	0.539	5.802
MergeSort	2501	80.0	0.981	0.537	4.568
MergeSort	2601	80.0	4.719	0.715	4.813
MergeSort	2701	80.0	0.979	0.481	5.993
MergeSort	2801	80.0	0.932	0.533	6.194
MergeSort	2901	80.0	0.712	0.448	5.760
MergeSort	3001	80.0	0.676	0.462	6.005
MergeSort	3101	80.0	0.716	0.650	7.564
MergeSort	3201	80.0	0.666	0.607	6.760
MergeSort	3301	80.0	0.667	0.267	8.944
MergeSort	3401	80.0	0.688	0.344	9.349
MergeSort	3501	80.0	0.828	0.252	10.791
MergeSort	3601	80.0	0.501	0.421	11.604
MergeSort	3701	80.0	0.421	0.230	8.784

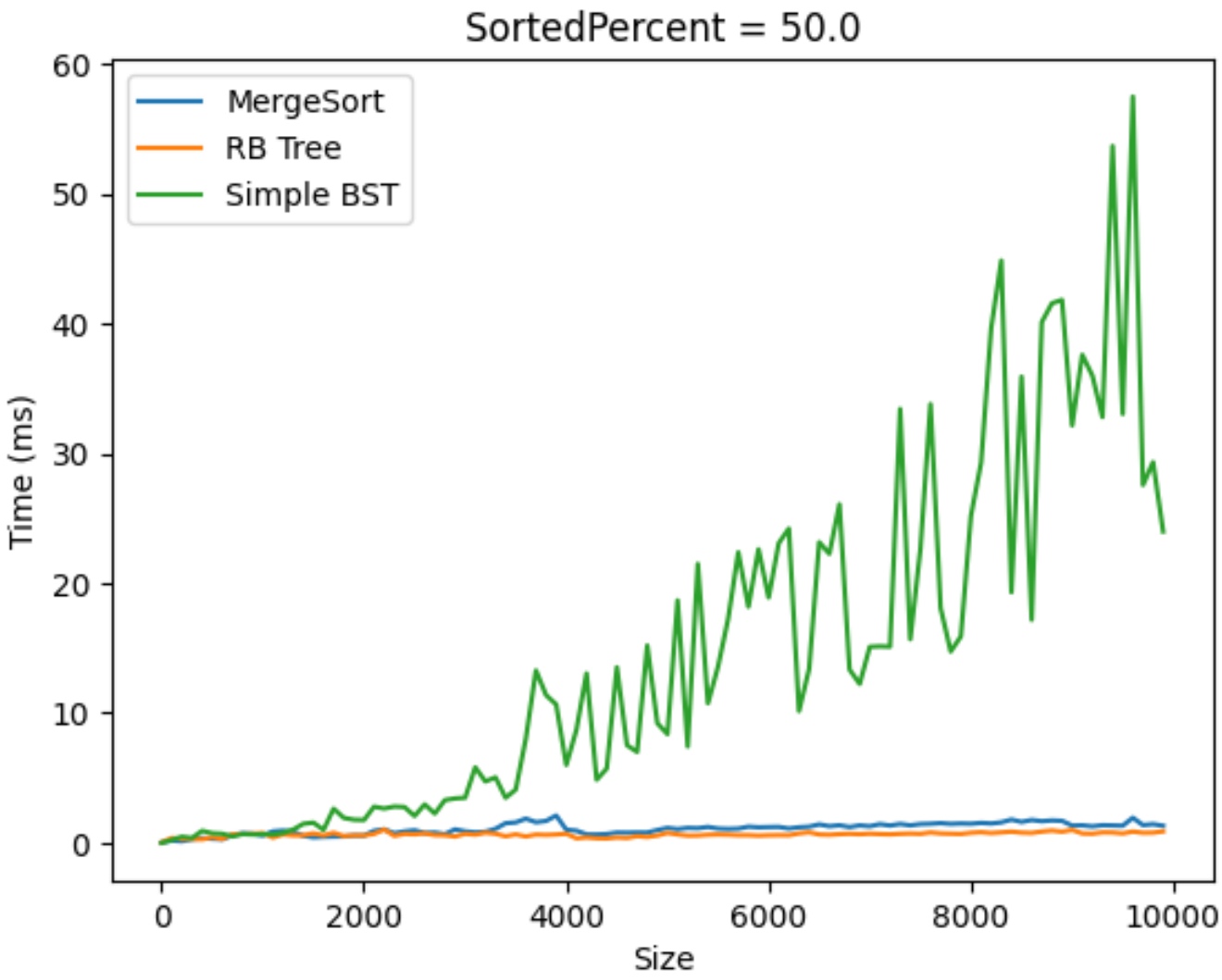
MergeSort	3801	80.0	0.476	0.262	10.092
MergeSort	3901	80.0	0.469	0.284	14.800
MergeSort	4001	80.0	0.448	0.266	13.109
MergeSort	4101	80.0	0.471	0.273	10.881
MergeSort	4201	80.0	0.474	0.262	15.817
MergeSort	4301	80.0	0.738	0.267	14.449
MergeSort	4401	80.0	0.498	0.291	15.767
MergeSort	4501	80.0	0.540	0.283	19.370
MergeSort	4601	80.0	0.554	0.289	19.592
MergeSort	4701	80.0	0.657	0.428	14.125
MergeSort	4801	80.0	0.559	0.315	20.509
MergeSort	4901	80.0	0.816	0.347	16.730

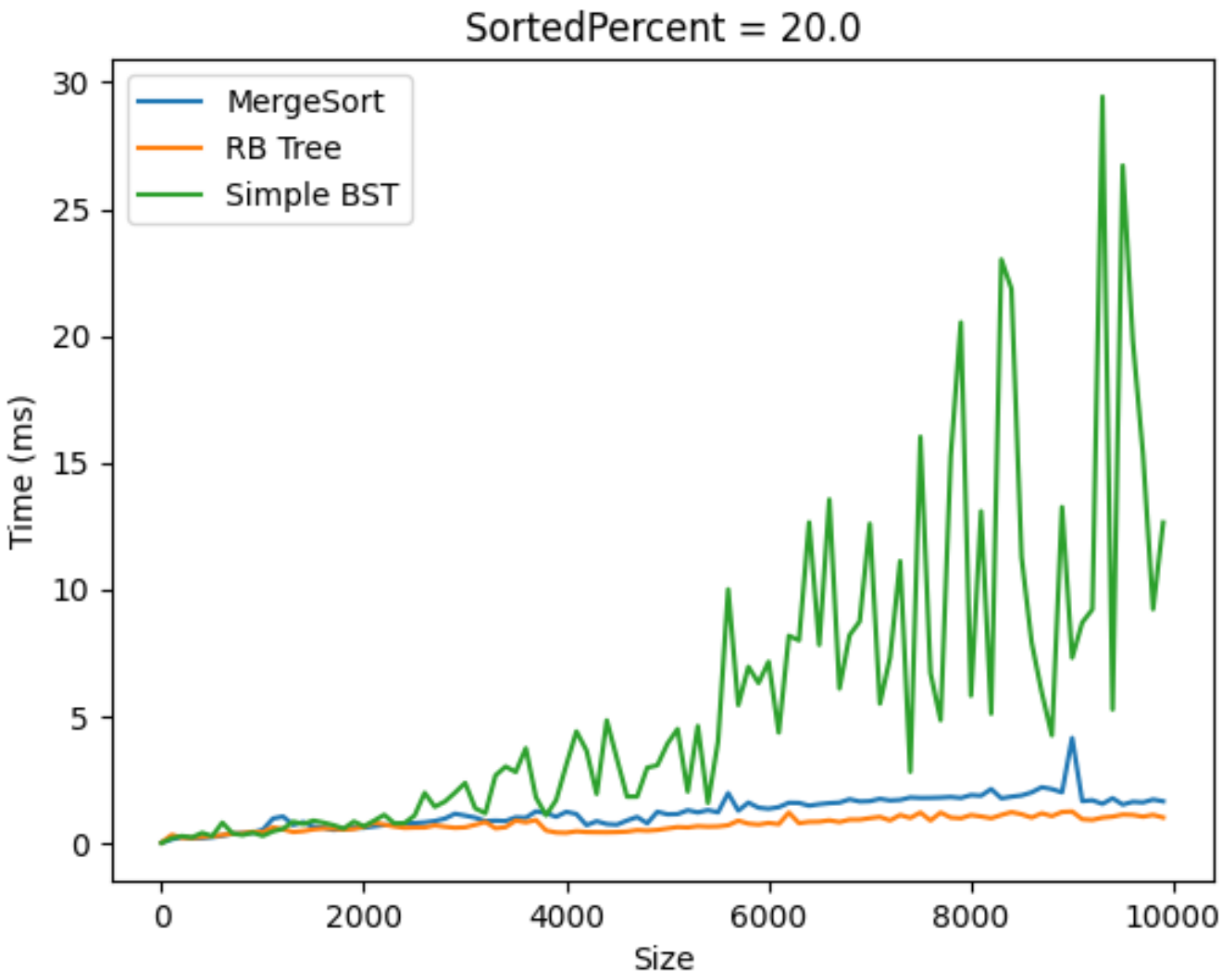
5 Regression Plots

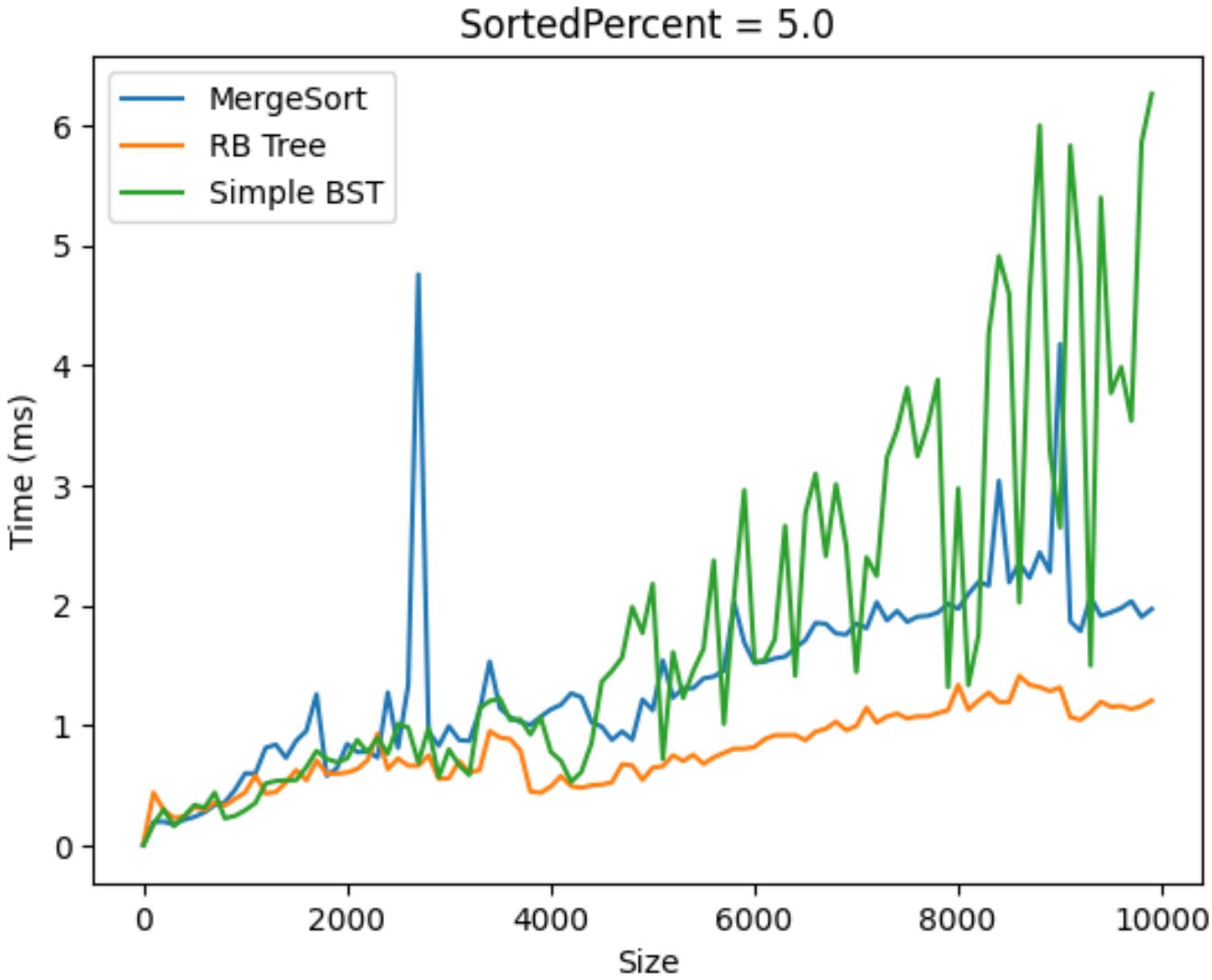

SortedPercent = 90.0
MergeSort
RB Tree
Simple BST
Time (ms)
0
20
40
60
80
0
2000
4000
6000
8000
10000
Size


SortedPercent = 80.0
MergeSort
RB Tree
Simple BST
Time (ms)
Size
0
10
20
30
40
50
60
70
0
2000
4000
6000
8000
10000


SortedPercent = 75.0
MergeSort
RB Tree
Simple BST
Time (ms)
Size
0
10
20
30
40
50
60
70
0
2000
4000
6000
8000
10000


SortedPercent = 50.0
MergeSort
RB Tree
Simple BST
Time (ms)
Size
0
10
20
30
40
50
60
0
2000
4000
6000
8000
10000


SortedPercent = 20.0
MergeSort
RB Tree
Simple BST
Time (ms)
Size
0
5
10
15
20
25
30
0
2000
4000
6000
8000
10000


SortedPercent = 5.0
MergeSort
RB Tree
Simple BST
Time (ms)
Size
0
1
2
3
4
5
6
0
2000
4000
6000
8000
10000